

Shopify Sales — Questions with Required DAX Formulas

Table names used below: `shopify_sales` (fact, line-item grain) and `Dim_Date` (calendar table you create). Column names match your file exactly, so these can be pasted almost as-is.

First, two structural measures/tables you need before any of the below will work correctly.

0. Date table (required for every time-based measure)

```
DAX

Dim_Date =
CALENDAR (
    MIN ( shopify_sales[Invoice Date] ),
    MAX ( shopify_sales[Invoice Date] )
)
```

Then in Power Query / model view: mark `Dim_Date` as a Date table (column = the date column), and relate `Dim_Date[Date]` → `shopify_sales[Invoice Date]` (1-to-many).

0.1 Order-level helper (avoids double counting, since your data is line-item grain)

```
DAX

Order Revenue Table =
SUMMARIZE (
    shopify_sales,
    shopify_sales[Order Number],
    "Order Total", SUM ( shopify_sales[Total Price Usd] ),
    "Order Date", MIN ( shopify_sales[Invoice Date] ),
    "Order Customer", MIN ( shopify_sales[Customer Id] )
)
```

Use this as a calculated table if you need true order counts / order-level AOV rather than line-item counts.

Stage 2 — Transaction Performance

Q1. What is Total Revenue?

```
DAX

Total Revenue = SUM ( shopify_sales[Total Price Usd] )
```

Q2. How many orders were placed (not line items)?

```
DAX

Total Orders = DISTINCTCOUNT ( shopify_sales[Order Number] )
```

Q3. What is Average Order Value (order-level, not line-item-level)?

```
DAX

AOV =
DIVIDE (
    [Total Revenue],
    [Total Orders]
)
```

Q4. How much tax was collected?

```
DAX

Total Tax Collected = SUM ( shopify_sales[Total Tax] )
```

Q5. How is revenue trending month over month?

```
DAX

Revenue MoM % =
VAR CurrMonth = [Total Revenue]
VAR PrevMonth =
    CALCULATE (
        [Total Revenue],
        DATEADD ( Dim_Date[Date], -1, MONTH )
    )
RETURN
DIVIDE ( CurrMonth - PrevMonth, PrevMonth )
```

Q6. Which Product Type drives the most revenue?

```
DAX

Revenue by Product Type = SUM ( shopify_sales[Total Price Usd] )
```

(Use as the value field in a bar chart with `shopify_sales[Product Type]` on axis — no extra DAX needed beyond Total Revenue, but the explicit measure keeps context clear if you also add rank logic below.)

Q7. Rank products by revenue (for a "top products" table/visual)

```
DAX

Product Revenue Rank =
RANKX (
    ALL ( shopify_sales[Product Type] ),
    [Total Revenue],
    ,
    DESC
)
```

Q8. What % of total revenue does each product contribute (Pareto view)?

```
DAX
```

```
% of Total Revenue =  
DIVIDE (  
    [Total Revenue],  
    CALCULATE ( [Total Revenue], ALL ( shopify_sales[Product Type] ) )  
)
```

Q9. What time of day / day of week do most orders happen?

```
DAX  
  
Orders by Weekday = DISTINCTCOUNT ( shopify_sales[Order Number] )
```

(Add a calculated column first: `Weekday Name = FORMAT(shopify_sales[Invoice Date], "dddd")`), then use it on the axis with the measure above.)

Q10. Average items per order

```
DAX  
  
Avg Items per Order =  
DIVIDE (  
    SUM ( shopify_sales[Quantity] ),  
    [Total Orders]  
)
```

Stage 3 — Customer Purchasing Behaviour

Q11. How many unique customers are there?

```
DAX  
  
Total Customers = DISTINCTCOUNT ( shopify_sales[Customer Id] )
```

Q12. First purchase date per customer (building block for new/returning logic)

```
DAX  
  
First Purchase Date =  
CALCULATE (  
    MIN ( shopify_sales[Invoice Date] ),  
    ALLEXCEPT ( shopify_sales, shopify_sales[Customer Id] )  
)
```

Add this as a **calculated column** on `shopify_sales`, not a measure — you need it row-by-row to classify orders.

Q13. Flag: is this row a New or Repeat purchase?

```
DAX
```

```

Order Type =
IF (
    shopify_sales[Invoice Date] = shopify_sales[First Purchase Date],
    "New",
    "Repeat"
)

```

(Calculated column, depends on Q12.)

Q14. Revenue from new vs. returning customers

```

DAX

Revenue - New Customers =
CALCULATE ( [Total Revenue], shopify_sales[Order Type] = "New" )

Revenue - Returning Customers =
CALCULATE ( [Total Revenue], shopify_sales[Order Type] = "Repeat" )

```

Q15. How many orders does the average customer place?

```

DAX

Avg Orders per Customer =
DIVIDE (
    [Total Orders],
    [Total Customers]
)

```

Q16. Total spend per customer (for ranking top customers)

```

DAX

Customer Total Spend = SUM ( shopify_sales[Total Price Usd] )

```

(Use in a table visual with **Customer Id** on rows — no rank needed unless you want a "Top N" card, in which case:)

Q17. Rank customers by spend

```

DAX

Customer Spend Rank =
RANKX (
    ALL ( shopify_sales[Customer Id] ),
    [Customer Total Spend],
    ,
    DESC
)

```

Q18. What % of revenue comes from the top 10% of customers?

```

DAX

```

```

Top 10% Customer Revenue % =
VAR CustomerCount = [Total Customers]
VAR Top10Count = ROUNDUP ( CustomerCount * 0.1, 0 )
VAR TopCustomers =
    TOPN (
        Top10Count,
        VALUES ( shopify_sales[Customer Id] ),
        [Customer Total Spend],
        DESC
    )
VAR TopRevenue =
    CALCULATE ( [Total Revenue], TopCustomers )
RETURN
DIVIDE ( TopRevenue, [Total Revenue] )

```

Stage 4 — Retention & Customer Lifetime Value

Q19. How many total orders has each customer placed? (for Frequency in RFM)

```

DAX

Customer Order Frequency =
CALCULATE (
    DISTINCTCOUNT ( shopify_sales[Order Number] ),
    ALLEXCEPT ( shopify_sales, shopify_sales[Customer Id] )
)

```

Q20. Repeat Purchase Rate — the % of customers who ordered more than once

```

DAX

Repeat Purchase Rate =
VAR RepeatCustomers =
    CALCULATE (
        DISTINCTCOUNT ( shopify_sales[Customer Id] ),
        FILTER (
            VALUES ( shopify_sales[Customer Id] ),
            CALCULATE ( DISTINCTCOUNT ( shopify_sales[Order Number] ) ) > 1
        )
    )
RETURN
DIVIDE ( RepeatCustomers, [Total Customers] )

```

Q21. Recency — days since each customer's last order (for RFM)

```

DAX

```

```

Recency (Days) =
VAR LastOrderDate =
    CALCULATE (
        MAX ( shopify_sales[Invoice Date] ),
        ALLEXCEPT ( shopify_sales, shopify_sales[Customer Id] )
    )
RETURN
DATEDIFF ( LastOrderDate, MAX ( shopify_sales[Invoice Date] ), DAY )

```

(Better built as a customer-level table measure; simplest done against Order Revenue Table from step 0.1 with TODAY() or a fixed report-refresh date instead of MAX(Invoice Date) for a live "days since" figure.)

Q22. Customer Lifetime Value (revenue-based, since no cost/margin field exists)

```

DAX

CLV (Revenue-Based) =
VAR AvgOrderValue = [AOV]
VAR AvgFrequency =
    DIVIDE ( [Total Orders], [Total Customers] )
VAR AvgLifespanMonths =
    DATEDIFF ( MIN ( shopify_sales[Invoice Date] ), MAX ( shopify_sales[Invoice Date] ), MONTH )
RETURN
AvgOrderValue * AvgFrequency * AvgLifespanMonths

```

State clearly in your report: this is **revenue-based CLV**, not profit-based, because cost/margin isn't in the dataset.

Q23. Gap between first and second purchase (re-engagement timing)

```

DAX

Days Between 1st and 2nd Purchase =
VAR CustomerOrders =
    CALCULATETABLE (
        VALUES ( shopify_sales[Invoice Date] ),
        ALLEXCEPT ( shopify_sales, shopify_sales[Customer Id] )
    )
VAR FirstDate = MINX ( CustomerOrders, shopify_sales[Invoice Date] )
VAR SecondDate =
    MINX (
        FILTER ( CustomerOrders, shopify_sales[Invoice Date] > FirstDate ),
        shopify_sales[Invoice Date]
    )
RETURN
DATEDIFF ( FirstDate, SecondDate, DAY )

```

(Best implemented against the Order Revenue Table from step 0.1 to avoid line-item duplication.)

Q24. Cohort month (month of first purchase, for cohort analysis)

DAX

```
Cohort Month =  
FORMAT ( shopify_sales[First Purchase Date], "YYYY-MM" )
```

(Calculated column, depends on Q12; use on rows of a cohort matrix visual against `Dim_Date[Month]` on columns, with `[Total Customers]` or a retained-customer measure as values.)

Q25. Customers retained N months after first purchase (cohort retention %)

DAX

```
Retained Customers =  
CALCULATE (  
    DISTINCTCOUNT ( shopify_sales[Customer Id] ),  
    FILTER (  
        shopify_sales,  
        shopify_sales[Invoice Date] > shopify_sales[First Purchase Date]  
    )  
)
```

Stage 3/4 — RFM Segmentation (combining the pieces above)

Q26. Simple RFM tier using Recency, Frequency, Monetary scores

DAX

```
RFM Segment =  
VAR R = [Recency (Days)]  
VAR F = [Customer Order Frequency]  
VAR M = [Customer Total Spend]  
RETURN  
SWITCH (  
    TRUE (),  
    F >= 3 && M >= 1000, "Champion",  
    F >= 2 && M >= 500,  "Loyal",  
    R <= 30 && F = 1,    "New / Recent",  
    R > 90,              "At Risk",  
    "Standard"  
)
```

Treat the thresholds (3, 1000, 500, 30, 90) as placeholders — replace with real quartile cut points from your data (use a Power Query or DAX quantile check first, don't hardcode blindly).

Stage 5 — KPI Cards for the Dashboard Summary

DAX

```
Total Revenue = SUM ( shopify_sales[Total Price Usd] )
Total Orders = DISTINCTCOUNT ( shopify_sales[Order Number] )
AOV = DIVIDE ( [Total Revenue], [Total Orders] )
Total Customers = DISTINCTCOUNT ( shopify_sales[Customer Id] )
Repeat Purchase Rate = [as defined in Q20]
CLV (Revenue-Based) = [as defined in Q22]
```

These six are the headline cards — everything else (product breakdown, RFM, cohort, trend) supports them.

Build Order (so formulas don't break on missing dependencies)

1. `Dim_Date` table + relationship
2. `First Purchase Date` calculated column → `Order Type` column → `Cohort Month` column
3. Core measures: Total Revenue, Total Orders, AOV, Total Customers
4. Dependent measures: Repeat Purchase Rate, Recency, Customer Order Frequency, CLV
5. Ranking/segmentation measures: Product Revenue Rank, Customer Spend Rank, RFM Segment
6. Cohort/retention matrix last, since it depends on everything above